

Pelayo González de Lena Rodríguez

Bioinformatician | Multi-omics · NGS · Mass
Spectrometry



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· Oviedo, Spain

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EXPERIENCE

FPI Severo Ochoa Research Fellow

Nov 2020 – Sep 2026

University of Oviedo · Plant Physiology Area, Dept. of Organisms and Systems
Biology

Including a research stay at the Spanish National Cancer Research
Centre (CNIO), Computational Cancer Genomics, and a contract at
CINN-CSIC within the European project eprObes.

Computational biologist

Oct 2019 – Nov 2020

GeoAI Analytics

GIS data analysis and drone flights with thermographic video.

Computer vision with OpenCV and machine learning applied to
geospatial data.

Bioinformatician

Jun – Oct 2019

Vivia Biotech, Tres Cantos (Madrid, Spain)

Bioinformatic analysis of *ex vivo* drug screening in leukaemia and other
haematological malignancies

Bioinformatician · external collaboration

2016 – 2018 · 2021 – 2022

Institute of Cellular and Molecular Studies (ICM) · Atrys Health, Lugo, Spain

Design of analysis pipelines for microarray and NGS technologies.

Contributed to the design of MeeApp, a direct-to-consumer genetic
testing application. Analysis on demand.

Research trainee · Molecular Oncology Unit

2016 – 2018

CIEMAT, Madrid, Spain

EDUCATION

PhD in Biology · University of Oviedo · defence expected

Sep – Oct 2026

MSc in Bioinformatics and Computational Biology · Carlos III Health Institute

2017

MSc in Biomolecules and Cell Dynamics · Autonomous University of Madrid

2016

BSc + MSc in Biology (5-year degree), Cell and Molecular Biology · University
of Oviedo

2011

Erasmus exchange · Università degli Studi di Napoli Federico II, Italy ·

2010 – 2011

Molecular and Cell Biology

Web programming and databases, 100 h · TalentUO, University of Oviedo

2025

BIOINFORMATICS SKILLS

Sequencing and arrays

End-to-end RNA-seq, from quality control to differential expression and functional enrichment · Nanopore long-read: basecalling, isoform analysis and RNA modification detection · lncRNA and non-coding RNA · expression and genotyping microarrays

Proteomics

Bottom-up histone LC-MS/MS with propionylation derivatisation · conversion from vendor formats · MS1 and MS2 spectra extraction · relative quantification of peptidofoms

Reproducibility

Docker · Snakemake · Git and GitHub · environment management · Linux and WSL2 · FAIR principles

Languages

Python · R and Bioconductor · MATLAB · Bash · SQL · JavaScript

Applied AI

Claude and Claude Code · OpenAI API · Model Context Protocol (MCP) · design of agents and skills for research · integration of language models into analysis workflows

Statistics

Differential expression · compositional data analysis · multivariate analysis and dimensionality reduction · machine learning · computer vision · experimental design and normalisation

SOFTWARE

epiprofile-plants — Extension of EpiProfile 2.0 for plant histone proteomics. MATLAB, GPL-3.0.

K-CHOPORE — Eight-stage Snakemake and Docker workflow for Nanopore direct RNA sequencing. Orchestrates 15 open-source tools from a single configuration file. MIT.

Code and projects: github.com/biopelayo · biopelayo.github.io

PUBLICATIONS

de Lena PG, Morales D, Escandón M, Meijón M, Valledor L, Cañal MJ, Pascual J. RNA sequencing platforms and bioinformatics tools. In: *Concepts and Strategies in Plant Sciences*, Springer Nature Singapore, 2026, pp. 31–65. [10.1007/978-981-95-5183-5_2](https://doi.org/10.1007/978-981-95-5183-5_2)

de Lena PG, Paz-Gallardo A, Paramio JM, García-Escudero R. Clusterization in head and neck squamous carcinomas based on lncRNA expression: molecular and clinical correlates. *Clinical Epigenetics* 2017;9:36. [10.1186/s13148-017-0334-6](https://doi.org/10.1186/s13148-017-0334-6)

de Lena PG, et al. Analysis of histone post-translational modifications in *Arabidopsis thaliana*. In: *Plant Proteomics: Methods and Protocols*. Methods in Molecular Biology, Springer, 2026. **Under review.**

de Lena PG. EpiProfile-PLANTS: extension of EpiProfile 2.0 for plant histone proteomics [software]. GPL-3.0. github.com/biopelayo/epiprofile-plants

TEACHING AND SUPERVISION

Python for genomic data analysis , 20 h, sole instructor · Asturian Institute of Public Administration	2025
Basic Course on Cancer Genomics , invited lecturer since the first edition · CIEMAT and CIBERONC	2017 – 2021
MSc thesis supervision : polygenic risk score pipeline for human complex diseases · Open University of Catalonia	2022
Linux, R and omics workflows · University of Oviedo	2019 – 2025
Programming and data analysis training · Oviedo City Council	2019 – 2025

COMPETITIVE FUNDING

FPI Severo Ochoa predoctoral contract · PRE2019-091395	Nov 2020 – May 2025
Spanish Ministry of Science and Innovation · University of Oviedo National competitive award for predoctoral training contracts. Five years of funding.	
Research contract · European project eprObes	Sep 2025 – Sep 2026
Horizon Europe, GA 101080219 · CINN-CSIC <i>Preventing lifetime obesity by early risk-factor identification, prognosis and intervention.</i> European consortium.	
Erasmus scholarship	2010 – 2011
European Union · Università degli Studi di Napoli Federico II, Italy Nine months in Molecular and Cell Biology, within the Biology degree at the University of Oviedo.	

LANGUAGES

Spanish · native speaker	
Italian · comprehension, interaction and speaking	C1
English · reading C1, remaining skills	B2
Catalan · listening B1, remaining skills	A2
Levels according to the Common European Framework of Reference.	